17-4

Stainless Steel 17-4 Product Guide

Applications | Specifications | Mechanical and Chemical Data | Tolerances

Overview

17-4 stainless has high tensile strength and hardness and excellent corrosion resistance up to high temperatures. It is typically used in gate valves, chemical processing equipment, pump shafts, gears, ball bearings, bushings, and fasteners.

Common Trade Names

UNS S17400, SAE type 630 stainless, AK Steel 17-4 PH®*, PH17-4, 17-4PH, 17Cr-4Ni, Precipitation Hardening

*AK Steel 17-4 PH® is a registered trademark of AK Steel Holding Corporation.

Other Resources

Safety Data Sheet | Weight Calculator | Mill Test Reports

Products

Purchase Round Bar 17-4 | Square Bar 17-4 | Sheet/Plate 17-4

Applications

Gate Valves
Chemical Processing Equipment
Pump Shafts
Gears
Ball Bearings
Bushings
Fasteners

Stainless 17-4 Specifications

ASTM	AMS	ASME
A564	5604	SA564
A693	5622	SA693
A705	5643	SA705
	5825	

Physical Properties 17-4

	Annealed	-H900
Density	0.280 lb/in3	0.282 lb/in3
Ultimate Tensile Strength	160 ksi	210 ksi
Yield Tensile Strength	145 ksi	200 ksi
Fatigue Strength	N/A	98 ksi
Shear Strength	N/A	120 ksi
Shear Modulus	N/A	11,000 ksi
Hardness Rockwell | Brinell	C36 | 363	C40-47 | 388-444
Elongation at Break Percentage	6-15%	12%
Reduction of Area	30-60%	35-40%
Modulus of Elasticity	N/A	28,000 ksi
Poisson's Ratio	0.28	0.28
Machinability Percentage	45%	N/A
Melting Point	2,550-2,620 °F	2,550-2,620 °F
Specific Heat	1.1 x 10^-1 BTU/lb-°F	1.1 x 10^-1 BTU/lb-°F
Thermal Conductivity	116 BTU-in/hr-ft^2-°F	116 BTU-in/hr-ft^2-°F
Electrical Conductivity	2.3% IACS	2.3% IACS

Chemistry Information

Element	Percentage
C	0.07 max
Cu	3 - 5
Cr	15 - 17.5
Fe	73
Mn	1 max
Nb + Ta	0.15 - 0.45
Nb	0.45 max
Ni	3 - 5

Chemistry Information

Element	Percentage
P	0.04 max
S	0.03 max
Si	1 max
Ta	0.45 max